Deep Learning for Sequences, Graphs, and Language Models

CS728 Lecture Plan and Resources

Spring 2026

Prerequisites

- Probability and statistics [Feller]
  - Random variables, entropy, mutual information
  - Discrete distributions: Bernoulli, multinomial
  - Continuous distributions: uniform, Gaussian, Beta, Dirichlet
  - Mean and variance
- ML [Bishop]
  - Supervised learning: naive Bayes, logistic regression, SVM
  - Unsupervised learning: k-means clustering, expectation maximization, singular value decomposition, principal component analysis
- Data structures and algorithms [CLRS]
  - List and graph manipulation
  - DFS, BFS, shortest path, max flow/min cut
  - Binary search, b-tree indexing, hashing
- CS635/CS6101 (topics repeated briefly are highlighted ♣ below)
  - Inverted index, TF, IDF, (sparse) vector space retrieval
  - Ranking evaluation: recall, precision, MRR, MAP, NDCG
  - Dense retrieval: single-vector DPR, multi-vector ColBERT, early cross-interaction

Course material

- [MS] Foundations of Statistical Natural Language Processing by Manning and Schutze.
- [G] Neural Network Methods for Natural Language Processing by Goldberg.
- [GNN] Graph Neural Networks: Foundations, Frontiers, and Applications.
- [TC] Introduction to Large Language Models by Chakraborty, NPTEL course.
- Papers will be linked to the lecture plan below.
- Past question papers (may be of limited utility).

Lecture Schedule

(Note that this looks excessively packed owing to the importation from CS772. We plan to thin down and clean up the topics within the first few days of the semester.)

Date	Day	Contents and resources
01-06	Tue	• Slides: course overview and logistics • What is (unstructured) language? What is (structured) knowledge? • Language as token sequence (what is a token? later) • Structured knowledge as tables, as feature-rich graphs • Discrete random variables, sequence models, Markov assumption • Lecture slides: 2026-01-06_TextAsTokenSequence.pptx
01-09	Fri	• q-gram language models, smoothing, back-off • Additive smoothing, Dirichlet prior • Prediction tasks: next token, label per token, passage-level label • How to evaluate a language model (perplexity) • Perplexity and its connection to cross entropy • ♣ Logistic regression and SVM with discrete tokens • Proof that the maximum entropy principle leads to logistic regression
01-13	Tue	• Slides: Text as token sequences; notes • Viterbi dynamic program and variations; limitations • BIO and BILOU state encodings • If labeled data is not enough, we can use Baum-Welch method • Word features, MEMM; limitation: local normalization
01-16	Fri	• Feature maps between inputs and candidate outputs • Global normalization, conditional random fields • ♣ SVMstruct (generating constraints on demand)
01-20	Tue	• Global normalization is more general than local • The bible on CRFs and semi-Markov models • Segment-CRF, PCFG, chart parsing
01-23	Fri	• Slides: Trees, graphs, message passing • Factor graphs and message passing • Probabilistic ML - Lecture 17 - Factor Graphs • Factor Graphs and message passing • Limitation of modeling language with hand-crafted token features
01-27	Tue	• ♣ Distributional vectors of words, clustering DVs • ♣ Latent semantic indexing / SVD over DVs • Slides: 2026-01-27_NeuralText.pptx
01-30	Fri	• ♣ Local context co-occurrence models: word2vec and GloVE • MEMM over word vectors = RNNs • Video: Stanford CS224N: NLP with Deep Learning | Winter 2019 | Lec…

02-03	Tue	• Training RNNs; exploding / vanishing gradients • Bi-LSTM/bi-GRU and bag-of-vectors encoding • ♣ Use of passage encoders in dense retrieval • Prediction task: sequence to sequence (translation, summarization) • Output distribution and sampling: beam search, nucleus sampling Additional resources • Training RNNs and bidirectional RNNs in [G], Ch-14 • Pascanu, Mikolov, Bengio on vanishing gradients in RNNs • LSTMs [Stanford NLP] • DPR and ColBERT for use of passage encoders in retrieval • Ilya Sutskever on seq2seq • Survey on language model decoding methods
02-06	Fri	• Fixed vector information bottleneck • Sequence to sequence with decoder-to-encoder attention • From decoder-to-encoder attention to self-attention • Lecture slides: 2026-02-06_Transformers.pptx Additional resources • Bahdanau, Cho and Bengio on the fixed-vector information bottleneck • Attention is all you need [all-to-all, causal stencils] • Rotary position embeddings for the transformer • Banded/sliding window stencil (Longformer) • More complex+random stencils (BigBird)
02-10	Tue	• Query, key, value; multiple attention heads • Feed-forward, layer normalize, residual connection → transformer • Positional embeddings, rotary embeddings • Attention stencils: all-to-all, causal, banded, random • Pretraining tasks: masked word reconstruction, next sentence classification, entailment classification
02-13	Fri	• Multiple heads, layer normalization • Encoder and decoder transformers • Encoder (BERT) pretraining • Decoder (GPT) pretraining • The Illustrated Transformer, microgpt
02-17	Tue	• 🔥 General LLM resources • Fine tuning BERT to specific tasks • Encoder-decoder (T5) pretraining
02-20	Fri	• 2026-02-18_FineTune.pptx • Parameter efficient fine tuning (PEFT) • Prompt and prefix tuning, instruction tuning • Low-rank model updates: LoRA and variations • Supporting long contexts; KVCache_notes.pdf
02-24	Tue	Midterm week

02-27	Fri	Midterm exam (30 marks) 17:00—19:00
03-03	Tue	Holi day
03-06	Fri	• Midterm paper review • Model pruning and distillation • Notes and simple sample code for distillation • 2026-03-06_InstructionTuning.pptx • Intro to reinforcement learning for LLMs
03-10	Tue	• Reinforcement learning • Agent (LLM), environment, action, state, reward • RLHF1, RLHF2
03-13	Fri	• Reinforcement learning • Policy gradient, actor-critic • RLHF3, RLHF4
03-17	Tue	• ACL 2023 RAG tutorial • Biasing a language generator via retrieval, retrieval as a tool • 2026-03-17_RAG_1.pptx
03-20	Fri	• Passage RAG, continued • REALM, RAG, FiD, Unlimiformer • Need for accessing graphs from LLMs • 2026-03-20_PassageRAG.mp4
03-24	Tue	• Two families of (knowledge) graph (KG) representations ○ Purely structural: TransE, DistMult, ComplEx, RotatE ○ Structure + local features: graph neural networks (GNNs) • 2026-04-01_KG.pptx
03-27	Fri	• Interpretability of language models (guest lecture, Anwoy Chatterjee) • 2026-03-27_Interpretability.mp4
03-31	Tue	Mahavir Jayanti
04-03	Fri	Good Friday
04-04	Sat	Written quiz (30 marks)
04-07	Tue	• 2026-04-07_GraphNeural_b.pptx • Interfacing graph representations to language models • KGQA applications • 2026-03-26_RAG_2.pptx
04-11	Sat	Guest lecture by Prof. Preethi • Speech + language models • Multilingual support • 2026-04-11_SpeechLanguage.mp4 • 2026-04-11_SpeechLanguage.pdf

04-??	???	• 2026-04-10_RAG_2.pptx • Table RAG with fine tuning ◦ 2d positional encoding for tables ◦ Attention modulation based on spatial cell relationships • Training-free approaches: convert tables to pretrained LLMs contexts • Structured output (e.g., code) generation, PICARD • Semantic interpretation, text2sql (guest lecture?) • RATSQL, DuoRAT, SMBoP, in-context text2sql
04-14	Tue	Ambedkar Jayanti
04-17	Fri	Guest lecture by Prof. Biplab • Visual language models (VLMs)
		Final exam
		Project and paper presentation
		Wishlist of topics we could not cover • Chain of thoughts • LM tokenization

Evaluation

- MCQ included in final exam = 10%
- Top two from three 30-mark pen-and-paper, no-device exams = 60%
- Weekly in-class online quizzes = 10–15%
- Programming assignments (in groups) = 20–30%
- Optional projects to substitute programming assignments or online quizzes
- Extra credit for presenting or critiquing papers